Topic 1.5 • Vectors and Motion in Two Dimensions

Notes (filled in)

Objective: split vectors into components and analyze projectile motion.

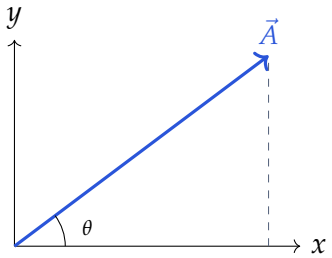
Components

A vector of magnitude A at angle θ above the $+x$ axis:

$A_x =$ $A\cos\theta$ $A_y =$ $A\sin\theta$

Back: $A =$ $\sqrt{A_x^2 + A_y^2}$ $\tan\theta =$ $\frac{A_y}{A_x}$

Example: $v = 10\,\text{m/s}$ at $37°$ ($\cos 37° \approx 0.8$, $\sin 37° \approx 0.6$): $v_x =$ $8\,\text{m/s}$, $v_y =$ $6\,\text{m/s}$.



Projectile motion

After launch (no air resistance) the two directions are independent :

Horizontal	**Vertical**
$a_x =$ 0: constant v_x	$a_y =$ $-g = -10\,\text{m/s}^2$
$\Delta x =$ $v_x t$	$\Delta y =$ $v_{0y}t - \frac{1}{2}gt^2$

The time is the same in both directions: find it from the vertical motion.

At the top of the path: $v_y =$ 0, but the acceleration is still g downward .

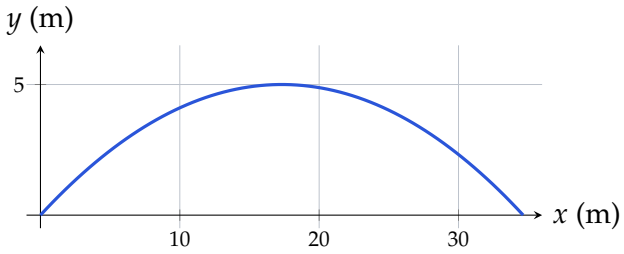
Example 1: horizontal launch

A ball rolls off a table 1.25 m high at 3 m/s.

Time to fall: $1.25 = \frac{1}{2}(10)t^2 \Rightarrow t =$ $0.5\,\text{s}$ Lands $3 \times 0.5 = 1.5\,\text{m}$ from the table.

A ball **dropped** from the same height lands at the same time .

Example 2: launch at an angle



$v_0 = 20\,\text{m/s}$ at $30°$: $v_{0x} =$ $17.3\,\text{m/s}$, $v_{0y} =$ $10\,\text{m/s}$

Time to the top: $10/10 = 1\,\text{s}$ Max height: $\frac{10^2}{2(10)} = 5\,\text{m}$

Total time (level ground): $2\,\text{s}$ Range: $17.3 \times 2 \approx 34.6\,\text{m}$

Summary

Split vectors into components with cos and sin. A projectile has constant horizontal velocity and a vertical acceleration of g downward; both motions share the same time. At the top, $v_y = 0$ but $a = g$.